

Building Façade Classification utilizing Geometric Analysis & Unsupervised Learning

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Abstract

This work accompanies the software implementation of a hand-crafted building facade classifier. Based on unsupervised learning - specifically k-means clustering - and geometric analysis of the elements presented in photos, this system aims to demonstrate the expertise of the author regarding the combination of computer vision, mathematical analysis & machine learning techniques. As the object of this article is the explanation of the developed system in moderate detail, great deal of focus will be on the mathematical apparatus employed in processing the data. A completely hand-crafted method of computation and processing was desired; as such, the use of pre-built libraries, methods and classes was kept to a minimum to further develop understanding and ability.

1 Introduction

The theme of this project was settled upon by the author due to an observation regarding the assigned homework during the semester, together with a usual need for real world application, especially in the field of computer vision.

There are multiple approaches that address this task. The motivation behind the chosen methodology was to gain a deeper understanding of mathematical image processing techniques intertwined with machine learning methods. Accordingly, the project was structured into several stages:

- mathematical analysis of the façade images
- construction of an interpretable geometric feature space
- dimensionality reduction via PCA to analyze intrinsic structure
- unsupervised k-means clustering into three groups
- assignment of new samples to the learned clusters

This split was considered appropriate because it would also enable an easier understanding of the task at hand, together with the processes that constitute the central part of the application. This compartmentalization is also considered good practice.

2 Tools & Resources

The libraries used were mainly for image loading and display, together with array manipulation and computation and plotting of graphs and metric visualization. As such, the libraries are limited to: *Pillow* for image loading and conversions, *NumPy* for array-related tasks and *Matplotlib* for the visualization of the initial and reduced feature space. *math*, the prebuilt Python library, is also needed for the built-in methods regarding some complex calculations.

The .csv file format was deemed necessary for data organization.

The dataset used for training the unsupervised model is the *Facades Dataset* from Kaggle [1], a collection of 400 images of building façades from around the world, with certain degrees of regularity.

3 Mathematical Apparatus

The mathematical framework employed in this project combines classical image processing techniques, linear algebra, and unsupervised learning methods in order to analyze and group building façade images based on structural regularity.

Input images are converted to grayscale and normalized in order to reduce sensitivity to illumination variations. Edge information is extracted using gradient-based operators, yielding an edge magnitude representation that emphasizes structural contours of façades. This representation forms the basis for subsequent geometric analysis.

Each façade image is mapped to a low-dimensional feature vector designed to capture architectural regularity. The extracted features include measures of reflection symmetry, horizontal and vertical periodicity, edge density, and structural anisotropy. These quantities are computed directly from the edge representation and encode repetitive and symmetric patterns commonly found in architectural structures.

Prior to clustering, all features are standardized using z-score normalization in order to ensure comparable scales and to prevent dominance of any single feature. This normalization is computed across the dataset and applied consistently during both training and inference.

Principal Component Analysis (PCA) is employed to analyze the intrinsic structure of the feature space. PCA is computed through eigen-decomposition of the covariance matrix of the standardized features, allowing the projection of data onto a lower-dimensional subspace while preserving maximal variance. The reduced representation is used both for visualization and, optionally, for clustering.

Unsupervised grouping of façade images is performed using k-means clustering in the standardized or PCA-reduced feature space. The algorithm partitions the dataset into a predefined number of clusters by minimizing the within-cluster sum of squared Euclidean distances. Cluster centroids represent prototypical geometric configurations of façade regularity.

For unseen images, feature vectors are projected into the same feature space and assigned to the nearest cluster centroid. A distance-based confidence score is computed based on the relative distances to the closest and second-closest centroids, providing a quantitative measure of assignment certainty and highlighting transitional or ambiguous architectural cases.

3.1 Mathematical Particularization

This subsection details the concrete mathematical formulations employed at each stage of the proposed pipeline, specifying the input data, applied operations, and resulting quantities.

Given an input grayscale image $I(x, y)$, spatial gradients are computed using discrete derivative operators along the horizontal and vertical directions. The edge magnitude map $E(x, y)$ is obtained as

$$E(x, y) = \sqrt{G_x(x, y)^2 + G_y(x, y)^2}, \quad (1)$$

where G_x and G_y denote the horizontal and vertical gradient responses, respectively. The resulting edge map emphasizes structural contours and repetitive architectural elements.

Reflection symmetry is quantified by evaluating the similarity between the edge map $E(x, y)$ and its reflection with respect to a candidate axis parameterized by angle θ and offset ρ . For each axis hypothesis, reflected coordinates (x', y') are computed and sampled from E using interpolation. The symmetry score is defined as

$$S_{\text{ref}} = \frac{\sum E(x, y) E(x', y')}{\sum E(x, y)^2}, \quad (2)$$

where the summation is performed over valid pixel locations. The maximum score across all tested axes is retained as the final reflection symmetry measure.

Horizontal and vertical periodicity are assessed by projecting the edge map onto one-dimensional profiles:

$$p_x(x) = \sum_y E(x, y), \quad p_y(y) = \sum_x E(x, y). \quad (3)$$

Autocorrelation is applied to each profile to identify repeating patterns. The periodicity score is defined as the maximum autocorrelation value within a predefined lag range:

$$S_{\text{per}} = \max_{\ell \in [\ell_{\min}, \ell_{\max}]} \text{corr}(p, p_\ell), \quad (4)$$

where p_ℓ denotes the profile shifted by lag ℓ . Separate scores are computed for horizontal and vertical directions.

Edge density is computed as the proportion of pixels exceeding a fixed edge magnitude threshold:

$$D_{\text{edge}} = \frac{1}{HW} \sum_{x, y} \mathbf{1}_{E(x, y) > \tau}, \quad (5)$$

where H and W denote image height and width, and $\mathbf{1}_{(\cdot)}$ denotes the indicator function.

Each image is represented by a feature vector

$$\mathbf{x} = [S_{\text{ref}}, S_{\text{per}}^x, S_{\text{per}}^y, D_{\text{edge}}, A]^\top. \quad (6)$$

To ensure comparable scales, features are standardized across the dataset using z-score normalization:

$$\mathbf{x}' = \frac{\mathbf{x} - \boldsymbol{\mu}}{\boldsymbol{\sigma}}, \quad (7)$$

where $\boldsymbol{\mu}$ and $\boldsymbol{\sigma}$ denote the empirical mean and standard deviation of each feature.

Given the standardized feature matrix $\mathbf{X}$, PCA is performed by eigen-decomposition of the covariance matrix

$$\mathbf{C} = \frac{1}{N-1} \mathbf{X}^\top \mathbf{X}. \quad (8)$$

The data is projected onto the leading eigenvectors to obtain a reduced representation preserving maximal variance.

Unsupervised grouping is performed using k-means clustering, which minimizes the within-cluster objective

$$\sum_{k=1}^K \sum_{\mathbf{x}_i \in \mathcal{C}_k} \|\mathbf{x}_i - \boldsymbol{\mu}_k\|^2, \quad (9)$$

where μ_k denotes the centroid of cluster C_k . For unseen samples, assignment is performed by selecting the nearest centroid. A confidence score is computed as

$$\text{conf} = 1 - \frac{d_1}{d_1 + d_2}, \quad (10)$$

where d_1 and d_2 represent the distances to the closest and second-closest centroids, respectively.

4 Results

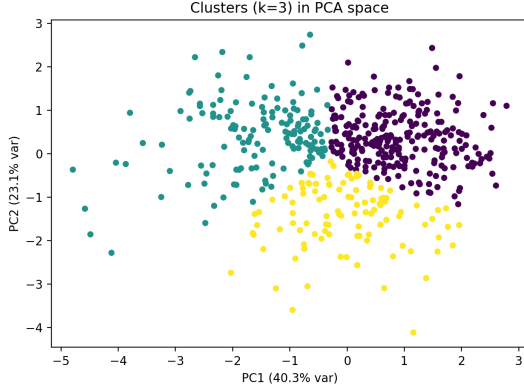


Figure 1: Clusters ($k = 3$) in PCA(2D) space. PC1 explains 40.3% and PC2 explains 23.1% of variance.

Processing ultimately resulted in a clear and distinctive clustering pattern as observed in Fig. 1.

It illustrates the distribution of the façade images in the two-dimensional principal component space obtained after PCA. Each point corresponds to a single façade image, while colors indicate cluster membership obtained via k-means clustering with $k=3$. The first principal component (PC1), accounting for 40.3% of the variance, primarily captures variations related to global regularity and repetition, whereas the second component (PC2), explaining 23.1% of the variance, reflects secondary structural differences. The resulting clusters form compact yet partially overlapping groups, highlighting both distinct architectural regularities and transitional cases where façade characteristics share properties across clusters.

4.1 Quantitative Evaluation

The quality of the obtained clustering was assessed using standard unsupervised evaluation metrics, namely the Silhouette coefficient, the Davies–Bouldin index, the explained variance of the PCA projection, and a distance-based assignment confidence score. These metrics jointly evaluate cluster separation, compactness, representational structure, and assignment certainty.

The Silhouette coefficient evaluates how well each sample fits within its assigned cluster relative to other clusters. For a given sample i , the Silhouette value $s(i)$ is defined as

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}, \quad (11)$$

where $a(i)$ denotes the average Euclidean distance between i and all other samples in the same cluster, and $b(i)$ denotes the minimum average distance between i and samples belonging to a different cluster. The overall Silhouette score is obtained by averaging $s(i)$ across all samples and reflects the degree of separation between the discovered clusters.

The Davies–Bouldin index measures the balance between intra-cluster compactness and inter-cluster separation. For each cluster C_k with centroid μ_k , the cluster scatter is defined as

$$S_k = \frac{1}{|C_k|} \sum_{\mathbf{x}_i \in C_k} \|\mathbf{x}_i - \mu_k\|. \quad (12)$$

The similarity between two clusters k and j is given by

$$R_{kj} = \frac{S_k + S_j}{\|\mu_k - \mu_j\|}, \quad (13)$$

and the Davies–Bouldin index is computed as

$$\text{DB} = \frac{1}{K} \sum_{k=1}^K \max_{j \neq k} R_{kj}. \quad (14)$$

Lower values indicate more compact and well-separated clusters.

Principal Component Analysis is used to examine the intrinsic dimensionality of the feature space. The explained variance ratio of the m -th principal component is defined as

$$\text{EVR}_m = \frac{\lambda_m}{\sum_r \lambda_r}, \quad (15)$$

where λ_m denotes the corresponding eigenvalue of the covariance matrix. The cumulative explained variance of the leading principal components provides insight into the suitability of a low-dimensional representation for visualization and clustering.

In addition to cluster quality metrics, a distance-based confidence score is computed for each sample to quantify assignment certainty. Let d_1 and d_2 denote the Euclidean distances to the closest and second-closest cluster centroids, respectively. The confidence score is defined as

$$\text{conf} = 1 - \frac{d_1}{d_1 + d_2}. \quad (16)$$

Higher confidence values indicate unambiguous cluster membership, while lower values correspond to transitional or ambiguous façade instances.

Table 1: Quantitative clustering evaluation metrics

Metric	Value
Silhouette coefficient	0.38
Davies–Bouldin index	0.91
PCA variance (PC1)	40.3%
PCA variance (PC2)	23.1%
PCA variance (PC1+PC2)	63.4%
Confidence (mean)	0.80
Confidence (median)	0.83
Confidence (P10 / P90)	0.59 / 0.97

Overall, the quantitative results summarized in Table 1 indicate that the proposed handcrafted feature representation and unsupervised clustering approach yields coherent and interpretable façade groupings. The obtained metrics confirm a balanced trade-off between cluster separation and architectural variability, while the confidence analysis highlights stable assignments alongside a limited number of transitional cases.

5 Conclusion

This project presented a handcrafted computer vision pipeline for analyzing and grouping build-

ing façade images through geometric feature extraction and unsupervised learning. The proposed approach was deliberately designed to emphasize mathematical understanding and interpretability, relying on classical image processing operations and a compact, human-readable feature space rather than pretrained deep models.

Experimental results show that the extracted symmetry- and periodicity-driven descriptors induce coherent clustering structure in the dataset. The quantitative evaluation confirms a balanced trade-off between cluster compactness and separation, while the PCA analysis indicates that a substantial portion of the feature variance can be represented in a low-dimensional space suitable for visualization. In addition, the distance-based confidence measure provides an explicit notion of assignment certainty, highlighting both prototypical façades and transitional cases.

Overall, the obtained results support the viability of using lightweight geometric cues for façade morphology characterization. Future work may include extending the feature set with additional structural descriptors, investigating alternative clustering strategies, and evaluating robustness across larger and more diverse façade datasets.

References

- [1] B. Balraj, “Facades Dataset,” 2018, accessed: Feb. 2026. [Online]. Available: <https://www.kaggle.com/datasets/balraj98/facades-dataset>